Def. Let V be a finite-dimensional inner product space over $F = \mathbb{C}$ or $\mathbb{R}$.

A linear operator $T: V \to V$ satisfying

$$\forall x \in V,\ \|T(x)\| = \|x\|$$

is called unitary when $F = \mathbb{C}$, orthogonal when $F = \mathbb{R}$.

Lemma. Let V be a inner product space, $H: V \to V$ be a Hermitian.

If for all $x \in V$, $\langle x, H(x)\rangle = 0$, then $H = 0$.

Proof. Let $x \in V$ be given. Then,

$$0 = \langle x + H(x), H(x + H(x))\rangle = \langle x + H(x), H(x) + H^2(x)\rangle$$
$$= \langle x, H(x)\rangle + \langle x, H^2(x)\rangle + \langle H(x), H(x)\rangle + \langle H(x), H^2(x)\rangle$$
$$= 0 + \|H(x)\| + \|H(x)\| + 0$$

Therefore, $H(x) = 0$.

Thm. Let V be a fin. dim. inn. V.S over $F = \mathbb{C}$ or $\mathbb{R}$.

For any linear operator $T: V \to V$,

1) For all $x \in V$, $\|T(x)\| = \|x\|$
2) For any $x, y \in V$, $\langle T(x), T(y)\rangle = \langle x, y\rangle$
3) $T \circ T^* = I$
4) $T^* \circ T = I$

Proof. suppose 1). we follow along 1) ⇒ 3) ⇒ 4) ⇒ 1).

Let $x \in V$ be given. Then,

$$\langle x, x\rangle = \langle T(x), T(x)\rangle = \langle x, (T^* \circ T)(x)\rangle \text{ which implies } \langle x, (I - T^* \circ T)(x)\rangle = 0.$$

And, since $(I - T \circ T^*)^* = I^* - (T \circ T^*)^* = I - T \circ T^*$, it is Hermitian,

therefore, $I = T^* \circ T$, by the Lemma. Clearly, this implies $I = T \circ T^*$.

Finally, if $T^* \circ T = I$, then for any $x \in V$,

$$\langle x, x\rangle = \langle (T^* \circ T)(x), x\rangle = \langle T(x), T(x)\rangle$$ ■

In $F = \mathbb{C}$.